PREP 4, LESSONS 7–8

Specific heat capacity

$\Delta E = m\,c\,\Delta\theta$; measuring specific heat capacity (required practical)

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Define internal energy. [1]

2 Define specific latent heat. [1]

3 Describe how to read a measuring cylinder accurately. [1]

Key ideas LESSONS 7–8

Show your working

4 2.0 kg of water is heated from 20 °C to 50 °C. Specific heat capacity of water = 4200 J/kg °C. Use $\Delta E = m\,c\,\Delta\theta$
Calculate the energy transferred to the water. [3]

5 The unit of specific heat capacity is | J/kg | J/kg °C | J/°C | [1]

6 A student heats a metal block with an immersion heater to find its specific heat capacity.

a Give the reason why the block is wrapped in insulation. [1]

b The student's value is higher than the true value. Explain why. [2]